Demostraciones clasificación

Machine Learning

Sea $\eta = \beta_0 + \beta^T x$. Queremos calcular la probabilidad p de que un suceso pase, $P(Y = 1)$, siendo $p \in (0, 1)$ (no tiene sentido calcular la probabilidad de un suceso que pasa nunca/siempre). Despejamos p de la ecuación:

$$log\left(\frac{p}{1-p}\right) = \eta \iff e^{log\left(\frac{p}{1-p}\right)} = e^{\eta} \iff \frac{p}{1-p} = e^{\eta} \iff p = e^{\eta}(1-p)$$

$$p = e^{\eta} - pe^{\eta} \iff p + pe^{\eta} = e^{\eta} \iff p(1 + e^{\eta}) = e^{\eta} \iff p = \frac{e^{\eta}}{1 + e^{\eta}}$$

Dividiendo toda la fracción por e^{η}

$$p = \frac{\frac{e^{\eta}}{e^{\eta}}}{\frac{1}{e^{\eta}} + \frac{e^{\eta}}{e^{\eta}}} = \frac{1}{\frac{1}{e^{\eta}} + 1} = \frac{1}{e^{-\eta} + 1}$$

Desarrollando η nos queda

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \equiv \sigma(\eta) \in (0, 1)$$

Propiedades:

- log es logaritmo neperiano, si no se indica lo contrario

- $e^{log(x)} = x$

- $\frac{1}{e^x} = e^{-x}$

- $log(x) = y \iff x = e^y$

- $\sigma(\eta) \in (0, 1)$ porque σ tiende a 0 cuando $x \to -\infty$, tiende a 1 cuando $x \to \infty$, es continua, y monótona creciente.

- $log(\frac{a}{b}) = log(a) - log(b)$

- $log(ab) = log(a) + log(b)$

- $log(a^b) = b \cdot log(a)$

Otras propiedades de la presentación:

1) $logit(1-p) = -logit(p)$

Recordemos que $logit(p) = log\left(\frac{p}{1-p}\right)$

$$logit(1-p) = log\left(\frac{1-p}{1-(1-p)}\right) = log\left(\frac{1-p}{p}\right) = log(1-p) - log(p)$$

$$-logit(p) = -log\left(\frac{p}{1-p}\right) = -\left(log(p) - log(1-p)\right) = log(1-p) - log(p)$$

cumpliéndose la igualdad.

2) $\sigma(-\eta) = 1 - \sigma(\eta)$

$$\sigma(-\eta) = \frac{1}{e^{-(-\eta)}+1} = \frac{1}{e^{\eta}+1}$$

$$1 - \sigma(\eta) = 1 - \frac{1}{e^{-\eta}+1} = \frac{(e^{-\eta}+1)-1}{e^{-\eta}+1} = \frac{e^{-\eta}}{e^{-\eta}+1} = \frac{1}{1+\frac{1}{e^{-\eta}}} = \frac{1}{1+e^{\eta}}$$

cumpliéndose la igualdad.

3) $\frac{\partial p(x)}{\partial x_j} = \beta_j p(x)(1-p(x))$, y en valor absoluto, $\leq \frac{|\beta_j|}{4}$

Sea $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$

Sea $p(x) = \frac{1}{1-e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_k x_k)}} = (1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_k x_k)})^{-1}$

$$\frac{\partial p(x)}{\partial x_j} = \frac{-1}{(1+e^{-\eta})^2} \cdot \frac{\partial(1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_k x_k)})}{\partial x_j} = \frac{-1}{(1+e^{-\eta})^2} \cdot (-\beta_j e^{-\eta}) = \frac{\beta_j e^{-\eta}}{1+e^{-2\eta}+2e^{-\eta}}$$

$$= \frac{\beta_j}{e^{\eta}(1+e^{-2\eta}+2e^{-\eta})} = \frac{\beta_j}{e^{\eta}+e^{-\eta}+2} = \frac{\beta_j}{e^{\eta}+e^{-\eta}+1+1} = \frac{\beta_j}{(e^{\eta}+1)(e^{-\eta})+1} = \frac{\beta_j}{e^{\eta}+1} \cdot p(x)$$

$$= \beta_j p(x)(1-p(x))$$

porque

$$1 - p(x) = 1 - \frac{1}{e^{-\eta}+1} = \frac{e^{-\eta}+1-1}{e^{-\eta}+1} = \frac{e^{-\eta}}{e^{-\eta}+1} = \frac{1}{e^{\eta}(e^{-\eta}+1)} = \frac{1}{1+e^{\eta}}$$

Sea $f(p) = p(1-p)$. Entonces, $f'(p) = 1-2p = 0 \iff p = \frac{1}{2}$. $p = \frac{1}{2}$ es máximo porque $f''(1/2) = -2 < 0$. Luego, $f(p) = p(1-p) \leq f(\frac{1}{2}) = \frac{1}{4}$.

Por tanto, tenemos $|\frac{\partial p(x)}{\partial x_j}| = |\beta_j p(x)(1-p(x))| \overset{(*)}{=} |\beta_j| p(x)(1-p(x)) \leq \frac{|\beta_j|}{4}$

(*) porque $p(1-p) \geq 0$ si $p \in [0,1]$